

End-Semester Examination
GPE201 (Earth and Planetary System)

Time: 10:00 - 13:00 hrs

Date: 11 May 2023
Maximum marks: 100

General instructions: There are SIX questions in total. Question No. 1 is compulsory. Out of the remaining FIVE questions, FOUR questions are to be answered. Parts of a question must be answered at the same place sequentially. The number of marks assigned to each question/part is indicated next to it. Neat sketches may be drawn to illustrate the answer, wherever required. Unless otherwise mentioned, symbols and notations have their usual standard meaning. Assume suitable data, if necessary, and indicate the same clearly.

Constants which may be needed:

Planck constant: $6.62607015 \times 10^{-34} \text{ J}\cdot\text{s}$
Stefan - Boltzmann constant: $5.670374419 \times 10^{-8} \text{ W}\cdot\text{m}^{-2}\cdot\text{K}^{-4}$
Constant B = $1.28 \times 10^{-11} \text{ W}\cdot\text{m}^{-2}\cdot\mu\text{m}^{-1}\cdot\text{K}^{-5}$
Wien's displacement constant = $2.8977685 \times 10^{-3} \text{ mK}$

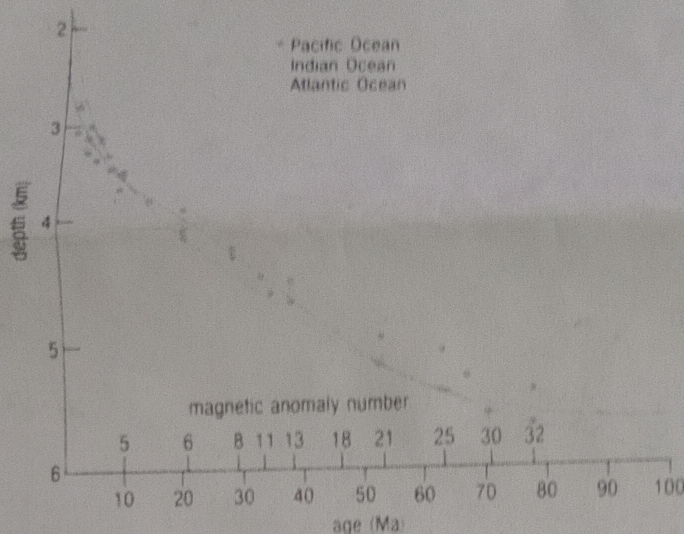
W/m²

- Q. 1(a) Are corals a net contributor to atmospheric CO₂? Justify your response.(5)
- Q. 1(b) How can climate change be distinguished between natural and anthropogenic forcers? *Internal wiggles.*(5)
- Q. 1(c) In Wien's law, what does λ_{max} signify? Calculate the maximal monochromatic exitance of isotherm of a black body at wavelength $\lambda_{\text{max}} = 1.184 \mu\text{m}$ at $T = 2,850 \text{ K}$?(2+3=5)
- Q. 1(d) What are the different forms of carbon prevalent in the ocean? Give details of DIC in the ocean water.(5)
- Q. 1(e) What is a Ferrel Cell? How is it connected with Hadley and Polar cells? *Dissolved inorganic Carbon*(5)
- Q. 1(f) What are isotones, isotonic, isobars, and isoelectronic? *Case (10)*(4)
- Q. 1(g) Which geophysical methods are primarily suitable for (i) mapping fossil fuels, (ii) delineating underground utilities, (iii) homeland defense, and (iv) engineering/construction site investigation?(4)
- Q. 1(h) Three water types (I, II, & III) of the Pacific Ocean mix to form a mixed water type IV. What are the proportions of water types I, II, and III in the mix water type IV and the T-S characteristics of water type IV? The characteristics of water types I, II, and III are listed below:(7)
- | Water Types | Temperature (°C) | Salinity (‰) |
|-------------|------------------|--------------|
| I | 5 | 35.0 |
| II | 3 | 34.0 |
| III | 1 | 35.5 |
- Q. 2(a) In the Keeling Curve, there is a clear upward trend in carbon dioxide concentrations and a smaller oscillating pattern in the data. Each year, CO₂ concentrations increase and decrease. What could be causing the annual cycle in carbon dioxide concentrations?(8)
- Q. 2(b) What are El Niño and La Niña? List the impacts of El Niño.(2+2+3=7)



Q.3

The image below demonstrates age-depth relationships across spreading ridges in the oceans. The solid line is a best-fit curve through observed points. The dashed line is a theoretical curve, calculated on the assumption that an increase of depth with age is due to the thermal contraction of the lithosphere as the plate cools on moving away from the spreading ridge axis.



Examine the above figure critically and respond to the following questions:

A 1000 m high seamount has its summit 3000 m below sea level. What might be the age of the seamount? Make suitable assumptions, if needed, and mention them clearly in arriving at the approximate age of the seamount. Do you think the seamount has a flat top? Justify your response. Are there any limitations in utilizing the aforementioned curve?

.....(6+5+4=15)

Q.4(a) What is the Earth system? Explain the major oceans of the hydrosphere.(2+6=8)

Q.4(b) Differentiate between continental crust and oceanic crust.(7)

Q.5(a) What is Rayleigh-Jeans law of electromagnetic radiation? Explain ultraviolet catastrophe.(4+4=8)

Q.5(b) What will be the useful spectrum corresponding to $T = 8,000 \text{ K}$ of a black body which emits EM radiation? In which part of the electromagnetic spectrum, the computed useful spectrum lies?(7)

Q.6 The following figure depicts major climate occurrences, labeled A to F as arrows, during the Cenozoic era. Identify them and explain any two climatic events.(6+5+4=15)

